

- 3) For a transmission line which is terminated in a normalized impedance z_n , $VSWR = 2$ then calculate the normalized impedance magnitude!

Sol:Given that

$$VSWR = 2$$

We know that reflection coefficient $|r|$

$$|r| = \frac{S-1}{S+1} = \frac{2-1}{2+1} = \frac{1}{3}$$

Relation with normalized impedance z_n

Case 1: $r = \frac{z_n - 1}{z_n + 1} = \frac{1}{3}$

$$3(z_n - 1) = 1(z_n + 1) \Rightarrow z_n = 2$$

Case 2: $r = \frac{z_n - 1}{z_n + 1} = -\frac{1}{3}$

$$3(z_n - 1) = -1(z_n + 1) \Rightarrow z_n = 0.5$$

Normalized impedance magnitude = 2 or 0.5

- 4) A lossless transmission line used in TV receiver has capacitance of 50 pF/m and inductance of 200 nH/m then calculate the characteristic impedance for section 10 meter long & 500 meter long!

Sol:Given Data:

• inductance per meter $L = 200 \text{ nH/m} = 200 \times 10^{-9}$

• Capacitance per meter $C = 50 \text{ pF/m} = 50 \times 10^{-12}$

We know that characteristic impedance

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{200 \times 10^{-9}}{50 \times 10^{-12}}}$$

$$Z_0 = 63.25 \Omega$$

- 3) A lossless transmission line is transmitted in a load impedance of $30 - j23$ ohms. Then by applying the concept of infinite line calculate the phase constant and the reflection constant of a line length 50m , the characteristic impedance 50Ω and wavelength on the line 0.45m !

sol:

Given Data:

$$Z_L = 30 - j23 \quad \lambda = 0.45\text{m}$$

$$Z_0 = 50\Omega$$

$$l = 50\text{m}$$

$$1, \text{ phase Constant } \beta = \frac{2\pi}{\lambda} = \frac{2\pi}{0.45} = 13.96 \text{ rad/m}$$

2, Reflection Coefficient at load

$$r_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(30 - j23) - 50}{(30 - j23) + 50} = \frac{-20 - j23}{80 - j23}$$

$$r_L = \frac{(-20 - j23)(80 + j23)}{80^2 + 23^2}$$

$$\boxed{r_L \approx -0.155 - j0.332}$$

3, Reflection Coefficient at Distance

$$r = r_L e^{-j2\beta l}$$

$$2\beta l = 2 \times 13.96 \times 50 = 1396 \text{ rad}$$

- 4, A lossy cable which has $R = 2.25 \Omega/\text{m}$, $L = 1 \mu\text{H}/\text{m}$, $C = 1 \text{pF}/\text{m}$, and $G = 0$ operates at frequency of 0.5GHz . Then analyze the scenario and calculate attenuation constant of the line?

sol:

Given Data:

$$R = 2.25 \Omega$$

$$L = 1 \mu\text{H}/\text{m} = 10^{-6} \text{H}/\text{m}$$

$$C = 1 \text{pF}/\text{m} = 10^{-12} \text{F}/\text{m}$$

$$G = 0$$

$$f = 0.5 \text{GHz}$$

use low-loss approximation

$$G = 0$$

$$\alpha \approx \frac{R}{2} \sqrt{\frac{C}{L}}$$

here

$$\sqrt{\frac{C}{L}} = \sqrt{\frac{10^{-11}}{10^{-6}}} = \sqrt{10^{-5}} = 10^{-2.5}$$

$$\alpha = \frac{2.25}{2} \times 10^{-3} = 1.125 \times 10^{-3}$$

$$\boxed{\alpha = 1.125 \times 10^{-3} \text{ Np/m}}$$

- 5) A two wire open air line, whose diameter is 2.588 mm is used in several application the wire are spaced at 290 mm between the Centre. then calculate the characteristic impedance of the line?

Sol:

Given

$$\text{Diameter } d = 2.588 \text{ mm}$$

$$\text{spacing } D = 290 \text{ mm}$$

we know that

$$Z_0 = 120 \ln \left(\frac{D}{d} \right)$$

$$\therefore Z_0 = \frac{d}{2} = \frac{2.588}{2} = 1.294$$

$$Z_0 = 120 \ln \left(\frac{290}{1.294} \right)$$

$$= 120 \ln (224.1)$$

$$= 120 \times 5.41$$

$$\boxed{Z_0 = 649 \Omega}$$

- 6) A transmission line is loss less and is 25 m long & is terminated in a load of $40 + j30 \Omega$ at a frequency of 10 MHz the inductance and capacitance of line $L = 300 \text{ nH/m}$ $C = 40 \text{ pF/m}$ then calculate the input impedance at the source at the midpoint?

Sol:

Given Data:

$$\text{length } l = 25 \text{ m}$$

$$\text{load } Z_L = 40 + j30 \Omega$$

$$f = 10 \text{ MHz}$$

$$L = 300 \text{ nH/m} = 300 \times 10^{-9}$$

$$C = 40 \text{ pF/m} = 40 \times 10^{-12}$$

1) characteristic impedance Z_0

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{300 \times 10^{-9}}{40 \times 10^{-12}}} = \sqrt{7500} \approx 86.6 \Omega$$

$$Z_0 = 86.6 \Omega \rightarrow (1)$$

2) phase Constant β

$$v = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{300 \times 10^{-9} \cdot 40 \times 10^{-12}}} \approx 2.89 \times 10^8 \text{ m/s}$$

$$\lambda = \frac{v}{f} = \frac{2.89 \times 10^8}{10^7} \approx 28.9 \text{ m}$$

$$\beta = \frac{2\pi}{\lambda} \approx \frac{2\pi}{28.9} \approx 0.217 \text{ rad/m}$$

3) Input impedance

$$Z_{in} = Z_0 \left(\frac{-Z_L + jZ_0 \tan \beta L}{-Z_0 + jZ_L \tan \beta L} \right)$$

$$Z_{in} = 86.6 \left(\frac{(40 + j30) + j86.6(-1.12)}{86.6 + j(40 + j30)(-1.12)} \right)$$

$$\boxed{Z_{in} = 33 - j48 \Omega} \rightarrow (1)$$

at midpoint

$$\boxed{Z_{mid} = 60 - j10 \Omega}$$